

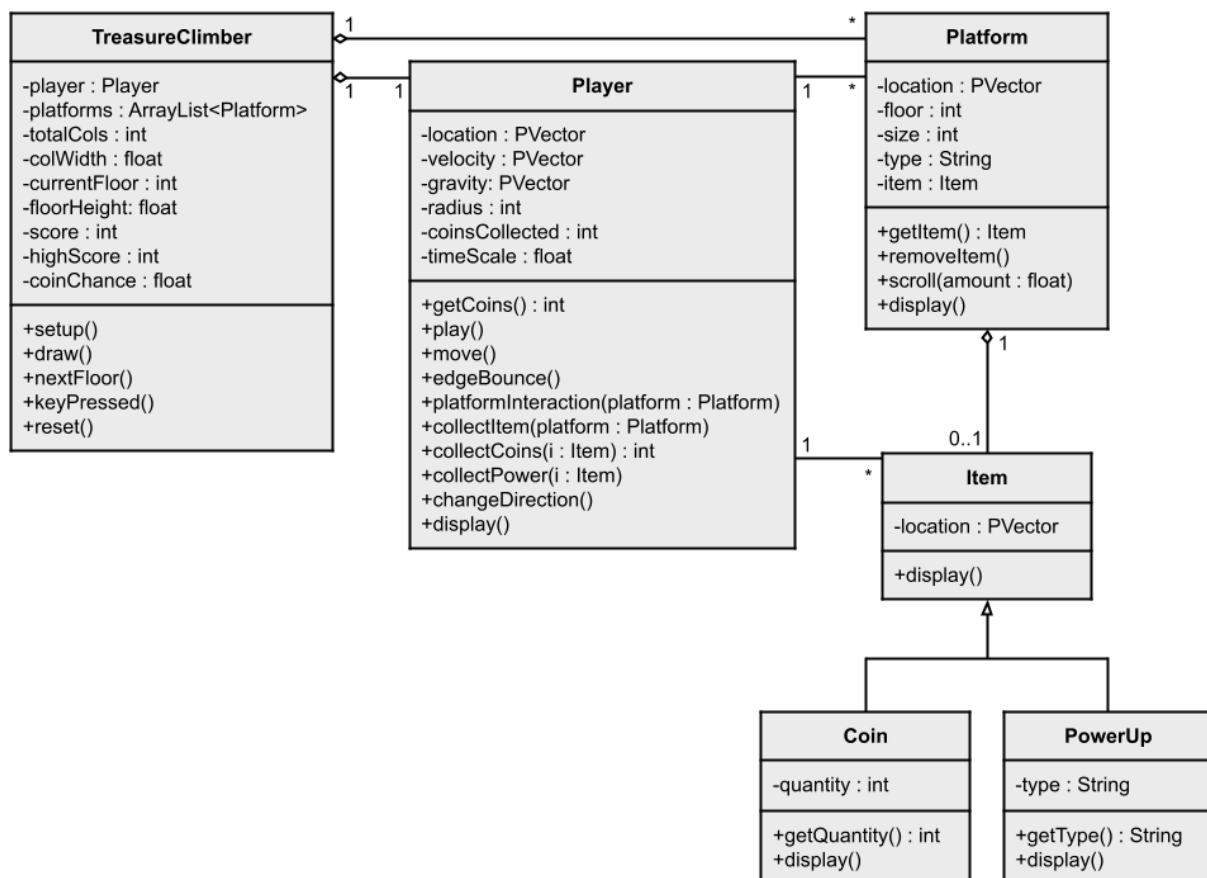
Intro

Period: 2
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Group: Mine Director
Project Title: Treasure Climber

Description

Treasure Climber is a game inspired by Doodle Jump. The player controls a character by having it move left and right to help it land on platforms to climb higher and higher. The player can also collect powerups that modify the game to make it easier. A final score will be calculated based on how high the player goes and how many coins the player collects. The game ends if the player falls below the bottom of the screen.

UML Diagram



Directions

The player spawns at the bottom of the mine and the character does not start until the player presses spacebar. Once the player starts, the character will keep moving in that direction until it hits a side of the screen or the player presses the spacebar again. As the player climbs higher, the character moves faster and faster, making the game progressively more difficult.

The objective of the game is to get the highest score possible. This can be achieved by climbing higher and collecting more coins. As of now, the game is planned to be endless.

Functionalities

Past Functionalities

- character with physics
 - left and right direction is swapped by pressing spacebar
 - keyPressed() in TreasureClimber, changeDirection() in Player
 - bounces off edges of the screen
 - bounces off platforms using Axis-Aligned Bounding Box (AABB) collision detection
 - starts immediately
- starting platforms of slightly different sizes arranged in seven columns
 - rectMode(CENTER) is used for locations
 - added totalColumns variable in TreasureClimber

New Functionalities

- player doesn't move until user presses spacebar (starts moving right)
- vertical scrolling and platform generation
 - platformBounce() → platformInteraction()
 - may separate these functionalities at a later time
 - colWidth, currentFloor, and floorHeight variables in TreasureClimber
- coins on platforms to collect
 - AABB collision detection
 - added Item and Coin classes
 - Item is an abstract class
 - location variable in Item
 - coinsCollected variable and collectItem() in Player
 - used instanceof and casting in player to access getQuantity(), a method that is in Coin but not in Item

- scoring system
 - score is the sum of the current floor and coins collected
 - highscore can always be seen in the top right corner of the screen
- powerups on platforms
 - AABB collision detection
 - jetpack: propels the player up (~12 floors)
 - timer: slows down the game (0.9x)
 - added timeScale variable in player

Future Functionalities

- more types of powerups
 - magnet: attracts coins
 - instantly collects all coins nearby OR
 - adds a force field (would require duration mechanic)
 - shield: grants extra life
- different platform types
 - trapdoor: disappears when landed on
 - spikes: ends game when landed on
 - conveyor: moves back and forth
- visuals
 - sprites for powerups (potentially for player and platforms as well)
- scenes
 - title screen
 - pause menu
 - game over menu
- sounds
 - upon collecting a coin or powerup
 - background music (settings menu)
- tutorial
 - likely a panel
- progression
 - less platforms the higher the player goes
 - player moves faster the higher the player goes
 - timer is for making the speed manageable
- optimizations and fixes
 - issues with platform spawns
 - old platforms are never deleted
 - entirely random; impossible jumps are possible
 - rare bug where player is propelled upwards indefinitely
 - making instance variables private and modifying methods to accomodate